

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

DIABETES

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

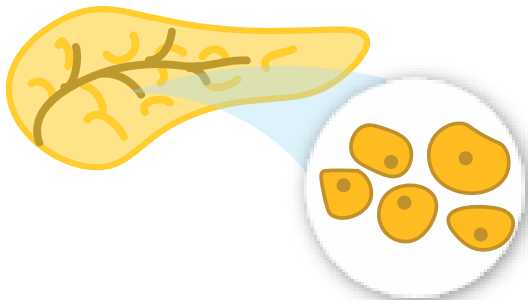
2hPG	2-hour Plasma Glucose
A1C	Glycated Hemoglobin A1c
ACEI	Angiotensin-Converting Enzyme Inhibitor
ACR	Albumin To Creatinine Ratio
ACS	Acute Coronary Syndrome
AEs	Adverse Effects
ASCVD	Atherosclerotic Cardiovascular Disease
AUD	Alcohol Use Disorder
BMI	Body Mass Index
BP	Blood Pressure
BG	Blood Glucose
CFG	Canada's Food Guide
CHF	Congestive Heart Failure
C/I	Contraindication
Cl-	Chloride
CrCl	Creatinine Clearance
CV	Cardiovascular
CVD	Cardiovascular Disease
D/I	Drug Interactions
DKA	Diabetes Ketoacidosis
DPP4i	Dipeptidyl Peptidase-4 Inhibitor
eGFR	Estimated Glomerular Filtration Rate
FPG	Fasting Plasma Glucose
GDM	Gestational Diabetes Mellitus
GI	Gastrointestinal
GIP	Glucose-dependent Insulinotropic Polypeptide
GLP1-RA	Glucagon-like Peptide-1 Receptor Agonist
H2O	Water
HDL	High Density Lipoprotein
HF	Heart Failure
IBD	Inflammatory Bowel Disease

ICU	Intensive Care Unit
K+	Potassium
LDL	Low Density Lipoprotein
LFT	Liver Function Test
LOC	Level Of Consciousness
LVEF	Left Ventricular Ejection Fraction
MACE	Major Atherosclerotic Cardiovascular Events
MAOI	Monoamine Oxidase Inhibitor
MENS2	Multiple Endocrine Neoplasia, Type II
MI	Myocardial Infarction
MTC	Medullary Thyroid Cancer
N/V/D	Nausea/Vomiting/Diarrhea
Na+	Sodium
OGTT	Oral Glucose Tolerance Test
PG	Plasma Glucose
PPAR-γ	Peroxisome Proliferator-activated Receptor Gamma
PPG	Post-prandial Glucose
Q	Every
RSV	Respiratory Syncytial Virus
S/E	Side Effect
SGLT2i	Sodium Glucose Co-transporter 2 Inhibitor
SJS	Stevens-Johnson Syndrome
SMBG	Self Monitored Blood Glucose
SU	Sulfonylurea
T1DM	Type 1 Diabetes Mellitus
T2DM	Type 2 Diabetes Mellitus
TENS	Transcutaneous Electrical Nerve Stimulation
TG	Triglyceride
TZDs	Thiazolidinediones
URI	Upper Respiratory Infection

PATHOPHYSIOLOGY

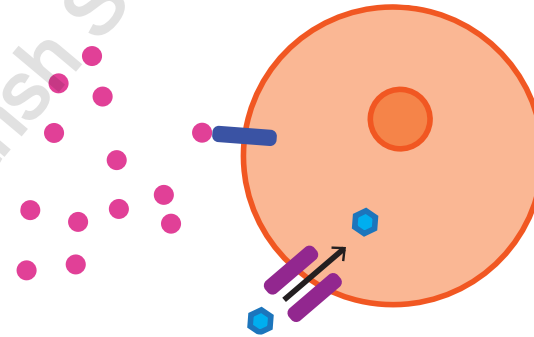
HEALTHY

Pancreas



Healthy beta cells

Insulin secretion



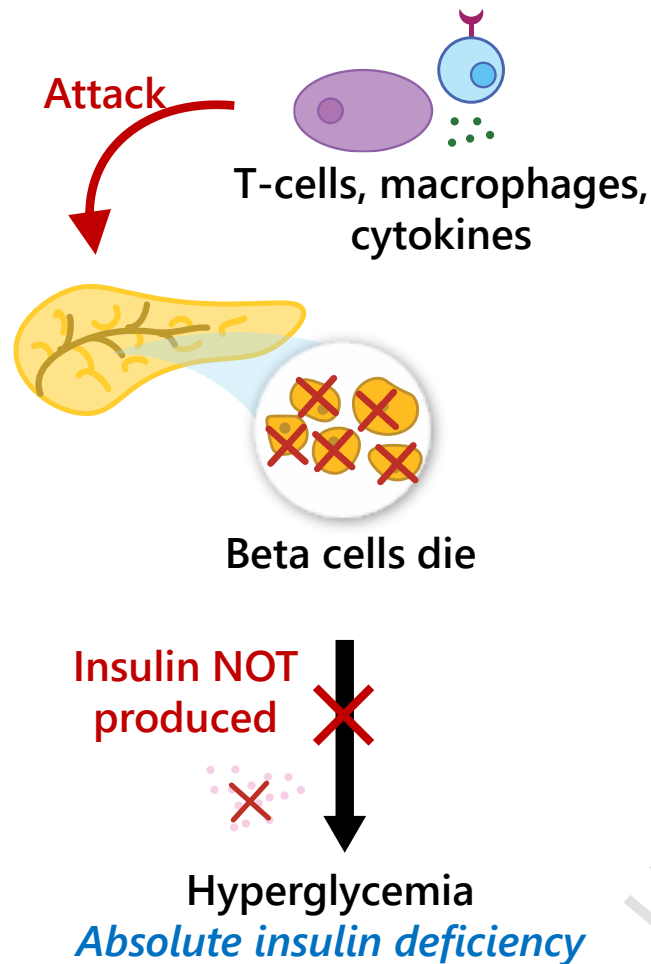
Insulin binds to the
Insulin receptor



Cells take
up glucose

PATHOPHYSIOLOGY

Type 1 Diabetes



DM has 2 types: **T1DM** & **T2DM**

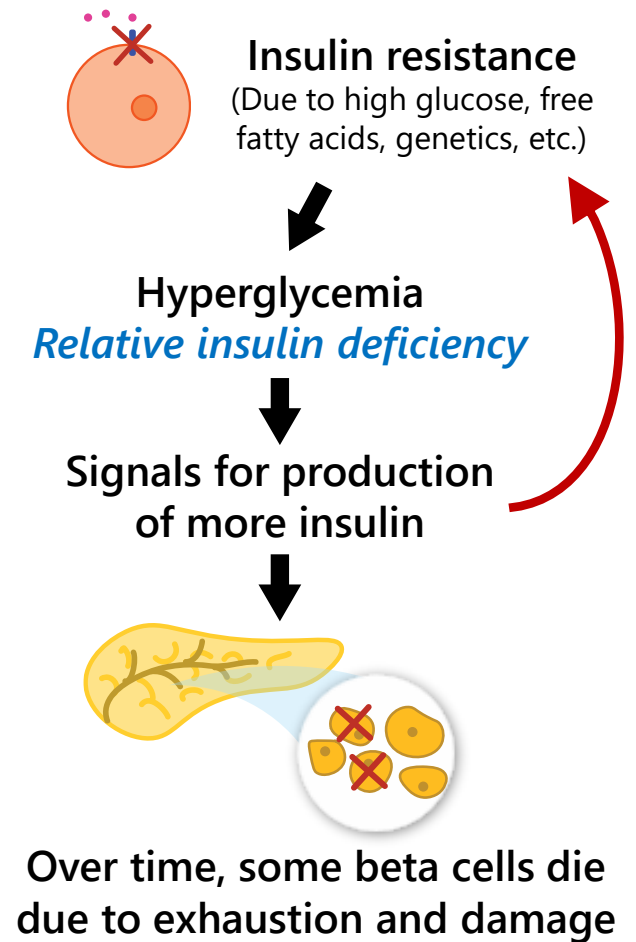
T1DM

Complete lack of insulin due to beta cell destruction in the pancreas; 5-10% of DM

T2DM

Relative lack of insulin due to multiple factors including hormone imbalance, excessive gluconeogenesis and insulin resistance; 90-95% of DM

Type 2 Diabetes



LOCK AND KEY ANALOGY

- **Insulin is the key** that unlocks a **cell's door** to allow **glucose to enter from the blood** to provide energy
- **Type I diabetes:** your body does not make its own insulin
 - No insulin "keys" → no ways to unlock the cell's door to allow glucose in → blood glucose level rises quickly
 - Treatment: insulin injection
- **Type II diabetes:** your body lacks enough insulin or has insulin resistance
 - Not enough insulin "keys" or key stops working as well → no ways to unlock the cell's door to allow glucose in → blood glucose level rises
 - Treatment: reduce sugar intake, increase body's ability to create insulin, or increase insulin sensitivity [and more...]



CLINICAL PRESENTATION

- Untreated diabetes results in hyperglycemia
- **Hyperglycemia** is characterized by the following:

Microvascular Complications

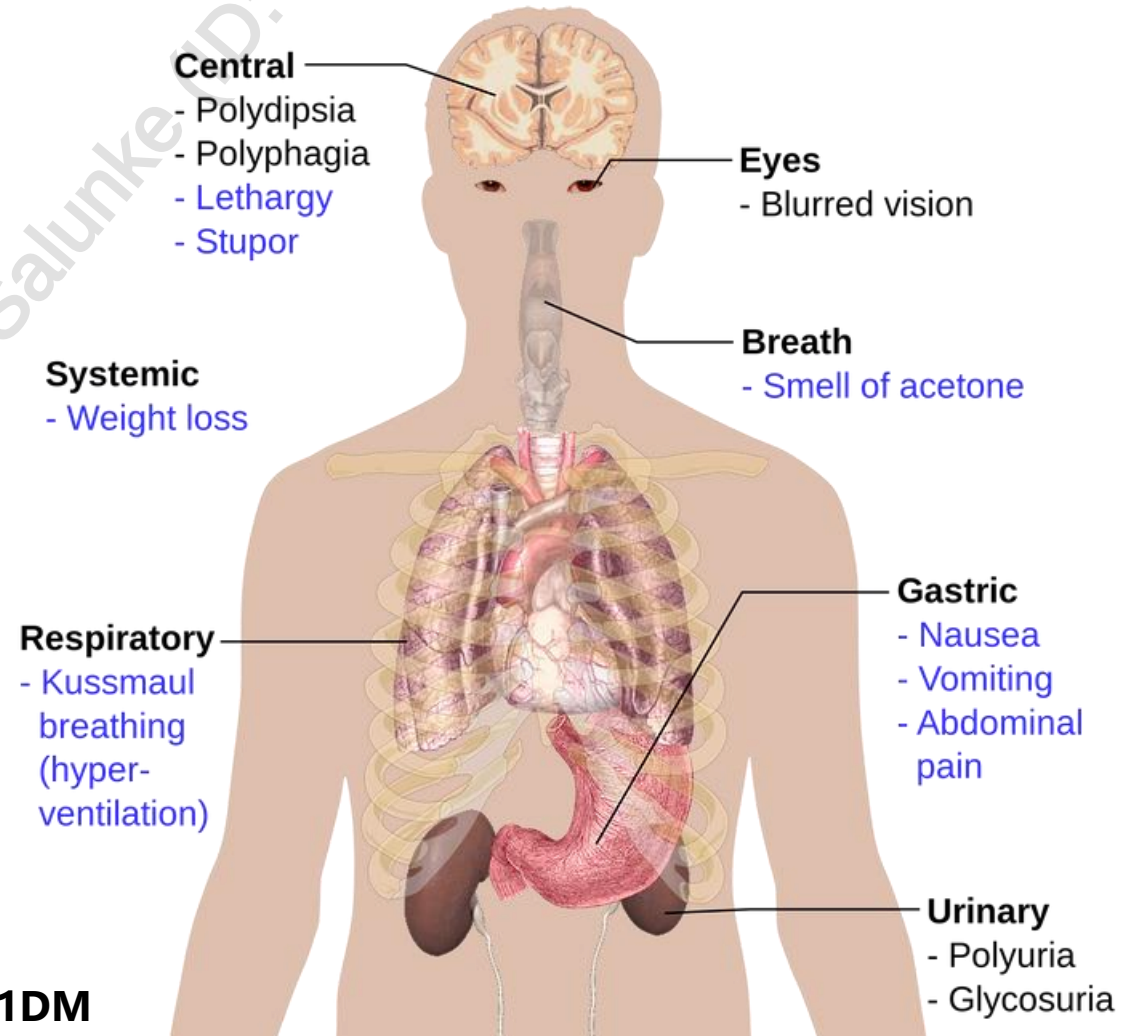
- Eye
 - Retinopathy
- Kidney
 - Nephropathy
- Nerves
 - Neuropathy

Macrovascular Complications

- Brain
 - Cerebrovascular disease
- Heart
 - Coronary artery disease
- Extremities
 - Peripheral vascular disease



Blue: More common in **T1DM**



DIAGNOSIS

FPG ≥ 7.0 mmol/L

Fasting = no caloric intake for at least 8 hours

OR

Random PG ≥ 11.1 mmol/L

Casual = any time of the day, without regard to the interval since the last meal

OR

2hPG in a 75-g OGTT ≥ 11.1 mmol/L

OR

HbA1c $\geq 6.5\%$ (in adults)

using a validated assay in the absence of factors that affect the accuracy of the A1C and **not for suspected T1DM**

- Must **repeat test to confirm** diagnosis (unless symptomatic hyperglycemia)
- **If possible T1DM, a confirmatory test should not delay initiation of treatment** to avoid rapid deterioration
- Ensure time of last meal known for FPG
- **HbA1c describes 3-month "average" of blood glucose**

DIAGNOSIS

Pre-diabetes

Refers to **ONE** of the following:

1. Impaired fasting glucose = 6.1 – 6.9 mmol/L
2. Impaired glucose tolerance: 2 hPG in 75 g OGTT 7.8 – 11.0 mmol/L
3. HbA1c of 6.0% – 6.4%

Gestational Diabetes

- Glucose intolerance with onset during pregnancy
 - If a patient has diabetes and then becomes pregnant, it is NOT gestational diabetes
- 5-10% of women with GDM ultimately have T2DM
- More common with:
 - 35 years+, obesity, family hx of DM, certain ethnic groups or with use of steroids

HYPOGLYCEMIA

Mild-to-Moderate

- **Mild: 3.0-3.9 mmol/L**
 - Autonomic symptoms
 - No neuroglycopenic symptoms or significant changes to mental status
 - **Individual can self-treat**
- **Moderate: <3.0 mmol/L**
 - Sometimes autonomic symptoms
 - Associated with neuroglycopenic symptoms
 - No significant changes to mental status
 - **Individual can self-treat**
- Treatment → "Rule of 15"
 1. Take **15 g PO of fast-acting carbohydrates** (4 glucose tabs or 3 tsp of table sugar or 1 tbsp of honey or 150 ml juice/soft drink)
 2. Check BG in 15 min
 - If **BG < 3.9**, then repeat treatment
 - if **BG > 3.9** and next meal is >1h away, encourage **15 g carbs + protein source**

Autonomic Symptoms

- Sweating
- Trembling
- Palpitations
- Anxiety
- Tingling
- Nausea
- Hunger

Neuroglycopenic Symptoms

- | | |
|------------------------------|-----------------------|
| • Reduced consciousness | • Vision changes |
| • Concentration difficulties | • Difficulty speaking |
| • Drowsiness | • Headache |
| • Confusion | • Dizziness |
| • Weakness | • Coma |
| | • Seizures |

HYPOGLYCEMIA

Severe

- **Severe: below normal, regardless of reading**
 - Associated with neuroglycopenic symptoms
 - Significant change to mental/physical status
 - **Requires assistance of another person to treat**
- Treatment for **patient able to swallow**:
 1. Take **20 g PO** of **fast-acting carbs**
 2. Check BG in 15 min
 - If **BG < 3.9**, retreat with 15 g fast-acting carbs
 - If **BG > 3.9** and next meal is >1 hour away, encourage snack with 15 g carbs + protein source
- Treatment for **patient unable to swallow**:
 - Outside of a healthcare setting: glucagon 1 mg SC/IM or 3mg intranasally
 - In a healthcare setting: 25 g IV dextrose

RISK FACTORS

T1DM (all non-modifiable):

- Family history
- Genetics
- Ethnicity (Finnish)
- Age (5-14)

T2DM

Modifiable:

- High body weight, excessive calorie intake, inactivity, other vascular risk factors

Non-modifiable:

- Age >40
- Ethnicity (SE Asians, Latin Americans, First Nations)
- Family history (first degree relative)
- Gestational diabetes, macrosomic infant

DRUG CAUSES (T2DM)

- Glucocorticoids
- Atypical antipsychotics
- Fluoroquinolones
- Thiazides
- Protease and Calcineurin Inhibitors
- Highly active antiretroviral therapy (HAART)
- Hyperglycemia-inducing agents

SECONDARY CAUSES (T2DM)

- Cancer
- Chronic pancreatitis
- Pheochromocytoma and Acromegaly
- Cushing's Syndrome
- Thyrotoxicosis and Malnutrition

GOALS OF THERAPY

- ✓ Achieve HbA1c target within 3-6 months of diagnosis
- ✓ Prevent hypoglycemia due to tight BG control
- ✓ Prevent macrovascular and microvascular complications
- ✓ Correct blood pressure and lipid abnormalities
- ✓ Educate patient regarding condition and needed lifestyle modifications
- ✓ Individualize therapy based on the patient characteristics and the BG levels

Measure	Target
FPG (mmol/L)	4.0-7.0 (to achieve A1C less $\leq 7\%$)
2 hr-post-prandial (mmol/L)	5.0-10.0 (if A1C is at target level) Otherwise, 5.0-8.0 if A1C not at target
HbA1c	$\leq 7\%$ for most patients $\leq 6.5\%$ in T2DM with low hypoglycemia risk (decreases CKD & retinopathy risks) Higher goals (7.1-8.5%) may be appropriate in certain patients such as the elderly, frail, patients with dementia, and those with hypoglycemia unawareness

NON-PHARMACOLOGICAL MEASURES



Self-management education

- Collaborative and interactive education and support
- Ensure patients can effectively and safely manage their disease by educating them on basic understanding of disease; role of diet, exercise, medications; self-monitoring of BG; management of sick days; hypoglycemia; care of feet; and awareness of other organs



Blood Glucose Monitoring

- Allows appropriate recognition of BG levels and provides immediate feedback about effects of therapy
- Self-monitoring of blood glucose & flash glucose monitoring



Ongoing Monitoring Of Other Organ Systems

- Detect comorbidities and complications
- Blood pressure, foot examinations, renal function, lipid profile, eye health

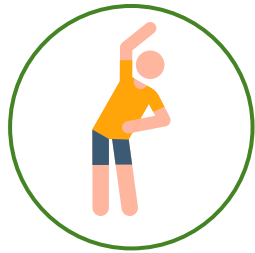
NON-PHARMACOLOGICAL MEASURES



Dietary Improvement/Nutrition Therapy

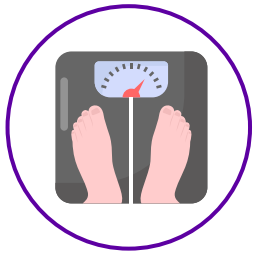
Leads to 1-2% reduction in HbA1c; Fats: 20-35% (saturated fats < 7%), Protein: 15-20%, Carbohydrate: 45-60%; Ethanol: ideally 2 standard drinks per week or less (although any amount of alcohol consumption is harmful)

- Follow CFG: Choose foods LOW in fat, salt, glycemic index and HIGH in fiber
- Patients on insulin require linking of carbohydrate intake and insulin dose



Exercise

- ≥ 150 min/week of moderate intensity aerobic exercise
- Regular resistance training 2 days per week



Weight/Abdominal Girth Modification

- Sustained weight loss of loss of $\geq 5\%$ of initial body weight improves glycemic control and CVD risk factors
- Achieved with healthy behaviour interventions
- Weight management medications can improve glycemic and metabolic control in patients with comorbid diabetes and obesity

PHARMACOLOGICAL MANAGEMENT

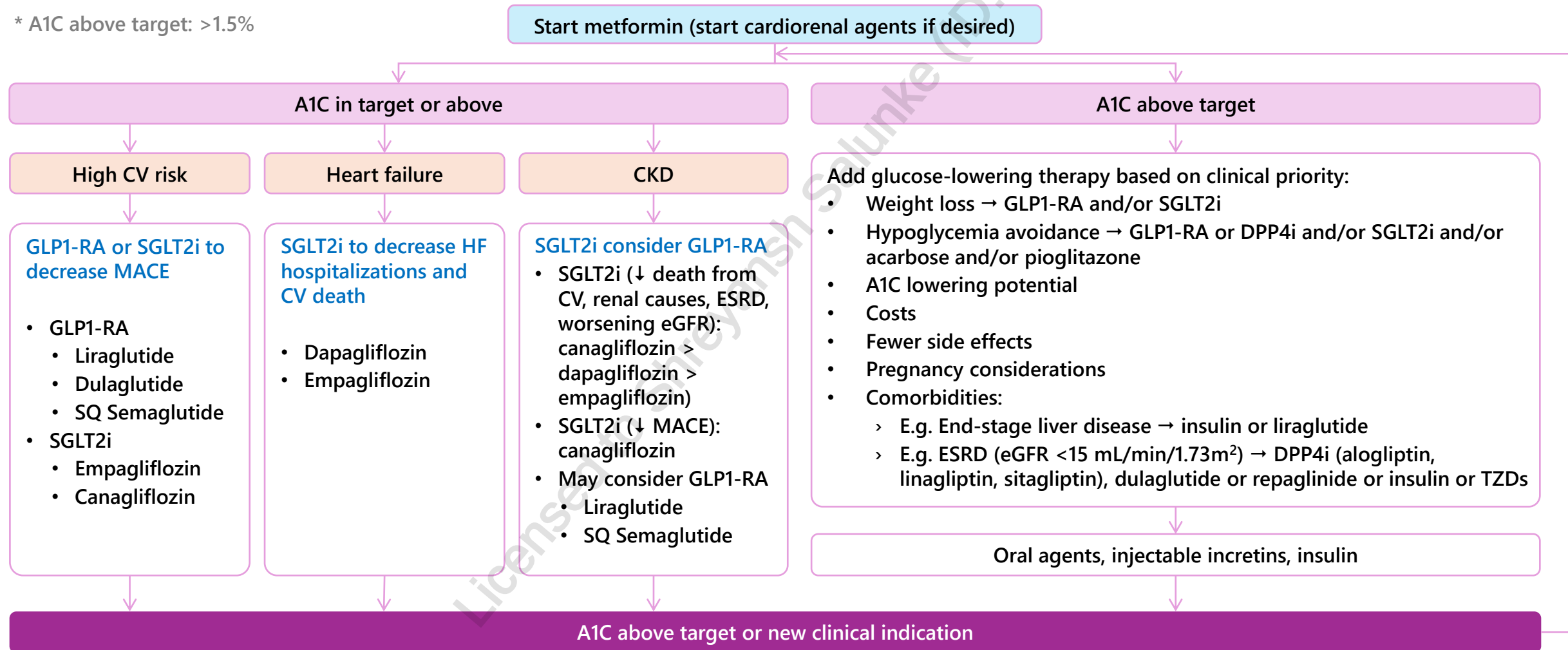
T2DM

- In select patients (e.g. HbA1C <1.5% away from goal)
 - Lifestyle changes (health behaviour modifications) acceptable initially
 - If A1C target is not reached within 3 months, start **metformin**
- Patients with metabolic decompensation
(*e.g. ketosis, unintended weight loss, polyuria, polydipsia, hyperglycemia*)
 - Start **insulin ± metformin**
 - Once stable, may taper/stop insulin and replace with other pharmacological agents
(*see next slide*)

PHARMACOLOGICAL MANAGEMENT

✓ Start non-pharmacological intervention (diet and exercise) in all patients and establish individualized A1C target

* A1C above target: >1.5%



PHARMACOLOGICAL MEASURES

BIGUANIDES & α -GLUCOSIDASE INHIBITORS

Drug class/A1C lowering	MOA	Safety	C/I and D/I
Biguanides • Metformin ~1%	• $\uparrow$ peripheral glucose uptake and insulin sensitivity • $\downarrow$ hepatic glucose production	• AEs: upset stomach, N/V/D, metallic taste, lactic acidosis, • B12 deficiency • Low risk of hypoglycemia • No weight gain	• C/I: eGFR <15-30 mL/min/1.73 m²*, severe hepatic failure (including AUD), severe dehydration, hx of lactic acidosis • Lower dose if eGFR < 45 mL/min/1.73 m²; monitor renal fxn and for signs of lactic acidosis • D/I: cimetidine & dolutegravir ($\uparrow$metformin)
α-Glucosidase Inhibitors • Acarbose ~0.7-0.8%	• Inhibit α-glucosidase enzyme in SI • Less glucose is absorbed since carbohydrates are not broken down	• AEs: flatulence, diarrhea, abdominal pain (dose related), hepatitis	• C/I: IBD, intestinal obstruction, intestinal diseases, ulceration, severe hepatic disease • $\downarrow$ digoxin levels • $\downarrow$ effect when used with digestive enzymes

* Metformin is contraindicated if eGFR is <15 mL/min/m², but this may vary based on condition e.g. in chronic kidney disease (CKD), if eGFR is <30 mL/min/1.73 m², metformin must be stopped at the time of or before the administration of iodinated contrast media due to the risk of lactic acidosis

PHARMACOLOGICAL MEASURES

INSULIN SECRETAGOGUES

Drug class	MOA	Safety	C/I and D/I
Sulfonylureas - Gliclazide - Glyburide ~0.6-1.2%	Depolarizes beta cells to stimulate insulin release	AEs: Hypoglycemia (higher with glyburide and eGFR < 60 mL/min) Weight gain (1.2 - 3.2 kg) - Headache - Dizziness - GI effects	C/I: T1DM, pregnancy, breastfeeding, ketoacidosis (with/without coma), severe liver/renal impairment, repaglinide with clopidogrel or gemfibrozil - Alcohol (↑ hypoglycemia) - β blockers (may mask hypoglycemia with sulfonylureas) - Drugs that cause hyper or hypoglycemia - CYP 3A4 Inhibitors/Inducers (meglitinides only)
Meglitinides Repaglinide ~0.7-1.1%		AEs: Hypoglycemia (lower than sulfonylureas) Weight gain (1.4 - 3.3 kg)	

PHARMACOLOGICAL MEASURES

THIAZOLIDINEDIONES & DPP-4 INHIBITORS

Drug class	MOA	Safety	C/I and D/I
Thiazolidinediones • Pioglitazone • Rosiglitazone ~0.5-1.5%	• ↑ PPAR-γ sensitivity to insulin • ↓ hepatic glucose production	AEs: • Worsening HF, fluid retention, weight gain (2 – 2.5 kg), fractures, macular edema (rare), URI, headache	• CYP 2C8 inhibitors/inducers (gemfibrozil - prolonged hypoglycemia) • Antihyperglycemic agents • C/I: HF, liver failure, bladder cancer with pioglitazone, MI with rosiglitazone – controversial
DPP-4 Inhibitors • Sitagliptin • Saxagliptin • Linagliptin • Alogliptin ~0.5-0.7%	• DPP-4 inhibition leads to ↑ in incretin levels	AEs: • Nasopharyngitis, hypersensitivity reactions • Rare: pancreatitis, severe joint pain, SJS	• Caution with history of pancreatitis or pancreatic cancer • CYP 3A4 interactions (except sitagliptin & alogliptin) • Avoid in HF patients (saxagliptin)

PHARMACOLOGICAL MEASURES

GLP-1 AGONISTS

Drug class	MOA	Safety	C/I and D/I
GLP-1 Agonists - Dulaglutide - Liraglutide - Lixisenatide - Semaglutide ~0.6 – 1.4%	↑ incretin concentrations	AEs: - N/V/D - Injection site reactions - Rare: acute pancreatitis, gallbladder disease - Weight loss (1.1 – 4 kg)	C/I: Pregnancy, personal/family hx of MTC or MENS2, caution with history of pancreatitis or pancreatic cancer ↓ gastric emptying (give agents requiring rapid GI absorption ≥ 1 hour before) - May need held prior to surgeries

GLP-1 AGONIST	Dosing Frequency
Liraglutide (with or without meals)	SC once daily
Dulaglutide (with or without meals)	SC weekly
Lixisenatide (prior to first meal of day)	SC once daily
Semaglutide (with or without meals)	SC weekly, PO daily

PHARMACOLOGICAL MEASURES

FYI

GLP-1/GIP RECEPTOR AGONIST

Drug class	MOA	Safety	C/I and D/I
GLP-1/GIP Receptor Agonist Tirzepatide	• ↑ incretin concentrations	AEs: • N/V/D • Injection site reactions • Rare: acute pancreatitis, gallbladder disease • Weight loss Other Safety: • Insulin and sulfonylurea doses may need decreased when starting tirzepatide • Do not use with DPP4i or GLP1-RA	• C/I: Pregnancy, breastfeeding personal/family hx of MTC or MENS2, caution with history of pancreatitis or pancreatic cancer • ↓ gastric emptying (give agents requiring rapid GI absorption ≥ 1 hour before) • May need held prior to surgery

Place in therapy is not well defined currently

PHARMACOLOGICAL MEASURES

SGLT2 INHIBITORS & INSULIN

Drug class	MOA	Safety	C/I and D/I
SGLT2 Inhibitors - Canagliflozin - Empagliflozin - Dapagliflozin ~0.5-0.7%	- Inhibits glucose reabsorption in the kidneys - Empagliflozin, canagliflozin and dapagliflozin slow nephropathy progression	AEs: - Genital mycotic infection (e.g. yeast infection), hypotension, hyperkalemia - Weight loss (2 – 3 kg) - rare DKA ↑risk of fractures and lower extremity amputations with canagliflozin in one trial - Reduced glycemic efficacy at lower eGFR	Renal Dosing (eGFR mL/min/1.73 m²): - Canagliflozin – if eGFR <60: reduce dose. If eGFR <30, do not start but may continue - Dapagliflozin – if eGFR <25: do not start but may continue - Empagliflozin – if eGFR <30: reduce dose. If eGFR <20: do not start but can continue - Canagliflozin: Strong enzyme inducers (e.g. carbamazepine, rifampin, phenytoin) - Loop diuretics ↑ risk of hypotension
Insulin ~0.9-1.2% or more	Acts like endogenous insulin in the body	AEs: Weight gain, hypoglycemia, lipohypertrophy (must rotate sites)	- Oral antidiabetic drugs - MAOI, βBs - Salicylates - Alcohol, Glucocorticosteroids

PHARMACOLOGICAL MEASURES

FYI

ORLISTAT

Drug class	MOA	Safety	C/I and D/I
Lipase Inhibitor • Orlistat (weight loss agent; may help reduce incidence of T2DM in at risk patients) ~0.2-0.4%	• Decreases fat breakdown and absorption by ~ 30% • Minimal systemic absorption	AEs: • Diarrhea, nausea, vomiting • Increased bowel movements • Flatulence • Steatorrhea • Abdo pain	• C/I in pregnancy, chronic malabsorption syndrome, cholestasis • ↑ INR (warfarin) • ↓ cyclosporine, levothyroxine (space 3-4 hours apart, respectively) • ↓ absorption of fat-soluble vitamins (separate by 2 hours or more) • ↓ anticonvulsants (monitor)

PHARMACOLOGICAL MEASURES

CLINICAL PEARLS

Metformin

- ↓ CV mortality → First line in T2DM
- ↓ insulin requirement (T2DM)
- Weight neutral
- Take with food to ↓ GI S/E (titrate to mitigate GI side effects),
- Hold for 48 hours if patient getting an x-ray/scan with an intravenous iodinated contrast material/dye
- Little benefit to increasing dose above 2g/day

Acarbose

- Weight neutral
- Do not use for marked hyperglycemia (weak efficacy)
- Take with each main meal (titrate to mitigate GI side effects)
- Monitor LFTs every 3 months during the first year, then periodically after

DPP-4 Inhibitors

- Weight neutral
- Alogliptin, saxagliptin and sitagliptin dose must be adjusted for renal impairment (linagliptin does not require renal dose adjustment, but must be used with caution if CrCl <15 mL/min)

PHARMACOLOGICAL MEASURES

CLINICAL PEARLS

Sulfonylureas

- Do not use glyburide in renal impairment
- Glipizide may be used with caution in eGFR 30-60 mL/min; contraindicated in severe renal impairment

Meglitinides

- Take 15-30 min prior to meals or right before. Delay or skip dose if meal is delayed or skipped
- Quicker onset compared to sulfonylureas and less risk of hypoglycemia

Thiazolidinediones

- Rosiglitazone warning: ↑ cardiovascular event risk; need written consent from patient
- ↓TG (pioglitazone), ↑HDL (both)

PHARMACOLOGICAL MEASURES

CLINICAL PEARLS

GLP-1 Receptor Agonists

- Can be used with other antidiabetic agents **except for DPP-4 inhibitors**
- **CV safety, but no proven cardiorenal benefit with exenatide ER, lixisenatide**
- Less A1C reduction with short acting agents
- **Monitor for retinopathy (especially if pre-existing) due to risk of progression with rapid ↓A1C**

SGLT2 Inhibitors

- Dose reduction required in renal dysfunction
- **Take before or with breakfast**
- Good foot care recommended, particularly in those with high-risk feet (loss of sensation, prior ulcer or amputation)

Insulin

- **Intensive therapy was shown to ↓ retinopathy, neuropathy, albuminuria & microalbuminuria**
- **First line in T1DM**
- **T2DM: Start insulin first line if symptomatic hyperglycemia with metabolic decompensation**

SUMMARY

Drug	HbA1c ↓	FBG	PPG	Hypoglycemic Risk	Weight Gain
Metformin	↓↓	↓	↓	-	-/↓
Sulfonylureas	↓↓	↓	↓	↑	↑
Meglitinides	↓ - ↓↓	↓	↓↓	-/↑	-/↑
TZDs	↓↓	↓	↓	-	↑↑
Acarbose	↓	↓	↓↓	-	-/↓
DPP-4 Inhibitors	↓	↓	↓↓	-	-/↓
GLP-1 Agonists	↓ - ↓↓	↓	↓ - ↓↓	-	↓
SGLT2 Inhibitors	↓	↓	↓↓	-	↓
Insulin	↓↓↓	↓↓	↓↓	↑↑	↑↑

Reduce A1C by 1-1.5%	Metformin, sulfonylureas, repaglinide, TZDs, GLP-1 analogues
Reduce A1C by ≤1%	Alpha-glucosidase inhibitors, DPP4 inhibitors, SGLT2 Inhibitors

TYPES OF INSULIN

✓ Mixed insulins are also available (combination of rapid OR short-acting + intermediate-acting insulin)

Insulin	RAPID-ACTING	REGULAR or SHORT-ACTING
	Lispro (Humalog®, Admelog®) Aspart (Novorapid®, Fiasp®) Glulisine (Apidra®)	Humulin-R® Novolin ge Toronto® Entuzity 500 units/mL®
Onset	10-15 mins (4 mins)	30 mins (15 mins)
Peak	60-120 mins (30-90 mins)	2-3 h (4-8 h)
Duration	3-5h (3-5 h)	6.5 h (17-24 h)
Basal/Bolus	Bolus	
Appearance	Clear	
Comments	• Less hypoglycemia than regular insulin • Take right before eating	• Take 30 min prior to eating • Regular insulin is available IV and used for hyperglycemic emergencies

TYPES OF INSULIN

✓ Mixed insulins are also available (combination of rapid OR short-acting + intermediate-acting insulin)

Insulin	INTERMEDIATE ACTING	LONG-ACTING	LONG-ACTING	ULTRA-LONG ACTING
	<i>NPH</i> Humulin N® Novolin ge NPH®	<i>Detemir</i> Levemir®	<i>Glargine</i> Lantus®, Basaglar® Toujeo® 300 units/mL	<i>Degludec</i> Tresiba®
Onset	1-3 h	90 mins		
Peak	5-8 h	No pronounced peak		
Duration	Up to 18 h	6-24 h	24h (> 30 h)	Up to 42 h
Basal/Bolus	Basal			
Appearance	Cloudy	Clear		
Comments	Given at bedtime or BID	- Peakless but detemir seems to have a shorter duration compared to glargine - Usually daily dosing, detemir may need to be given BID at lower doses		

TYPES OF INSULIN

FYI

✔ Once weekly insulin was just released to the market in June 2024 and is not in guidelines yet.

Insulin	ONCE WEEKLY <i>Isulin icodec</i> <i>Awigli®</i>
Onset	Unknown
Peak	15-18 h
Duration	Approximately one week
Basal/Bolus	Basal
Appearance	Clear and colourless solution
Comments	- Starting dose: 70 units once weekly OR previous basal dose (rounded to nearest 10) x 7 - Titrate by 20 units weekly (depending on patient factors)

PREMIXED INSULINS

FYI

Type of Premixed Insulin	Trade Name/Components
Rapid-acting insulin + intermediate-acting insulin	Humalog® Mix 25 (25% lispro, 75% lispro protamine) Humalog® Mix 50 (50% lispro, 50% lispro protamine) Novomix® 30 (30% aspart, 75% aspart protamine)
Short-acting insulin + intermediate-acting insulin	Humulin® 30/70 (30% regular, 70% NPH) Novolin® ge 30/70 (30% regular, 70% NPH)

INSULIN SWITCHING

Original Insulin (Stop)	New Insulin (Start)	Conversion
RA	RA	1:1
SA	RA	1:1
RA	SA	10-20% reduction
NPH once daily	Glargine, detemir	1:1
NPH twice daily	Glargine, detemir	20% dose reduction given once daily
Glargine, detemir	NPH once or twice daily	20% dose reduction
Toujeo®	Lantus	20% dose reduction
Tresiba®	Lantus	20% dose reduction
Entuzity	Others	Should NOT do

INSULIN DOSING

Bolus insulin

- Covers meal related increase in BG
- Rapid acting insulin preferred over regular insulin

T1DM

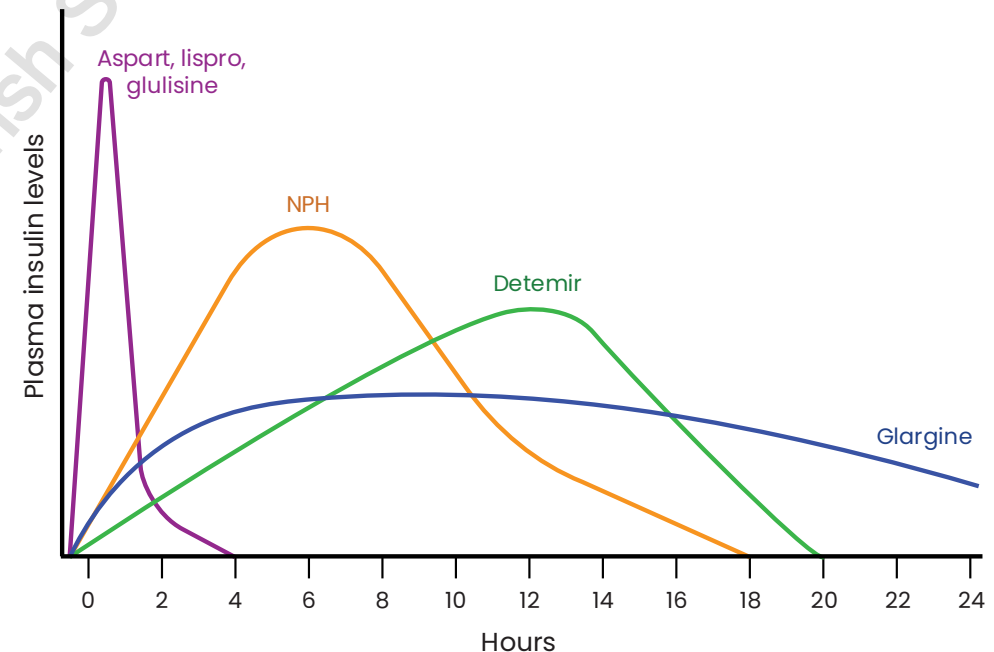
- Bolus + basal insulin required
- Many different types of insulin-start regimens available
- Many patients utilize insulin pumps

T2DM

- Basal insulin at bedtime can be started if BG cannot be controlled with oral agents
- Monitor hypoglycemia patterns

Basal insulin

- Covers BG throughout the day (non-meal related)
- Peak-less insulin preferred over NPH



INSULIN TITRATIONS

- Don't adjust insulin dose based on a single value, look at the **average values for a minimum of 3 days** to determine whether insulin dose should be changed
- **Hypoglycemia should be corrected before hyperglycemia** due to safety concerns
- Insulin doses are usually adjusted by **10% (1-2 units)**
- If all the average blood glucose values are high, target the **first elevated blood glucose of the day**

INITIATING INSULIN IN T2DM PATIENT

1. Start **basal** insulin (glargine, detemir, degludec) at a dose of **0.1-0.2 units/kg/day** (usually approx. 10 U QHS) and titrate 1 unit/day to AM FBG target (4-7 mmol/L)
 - Stop increase if there is any episodes of nocturnal hypoglycemia or the patient has 2 hypoglycemic episodes in 1 week
2. If glycemic targets have not been met, add one dose of bolus insulin per day (when patient typically has largest meal)
3. If glycemic targets have not been met, add multiple doses of bolus insulin per day at all meals

SIDE EFFECT COUNSELLING

Most common side effects:

**weight gain and
lipohypertrophy**



**Avoid lipohypertrophy by
rotating injection sites**

PHARMACOLOGICAL MANAGEMENT

T2DM

Initiating Insulin

(symptomatic hyperglycemia, metabolic decompensation or oral antihyperglycemics insufficient)



NEVER use GLPRA with DPP4I due to similar MOA

Start basal insulin: 0.1 – 0.2 units/kg/day, titrate 1 unit/day to target CBG

(if on SUs, meglitinides or TZDs, consider stopping or adjusting due to hypoglycemia risk)

Re-assess in 3-6 months and advance therapy as needed

Consider SGLT2i or [GLP1 RA or DPP4I] **sequentially** or together

Note: if eGFR >45 add SGLT2i for glycemia, otherwise, may add SGLT2i if eGFR >30 for cardiorenal benefit

Re-assess in 3-6 months and advance therapy as needed

Step-wise bolus insulin. Starting with 1 meal/day and increase to each meal as needed

Example starting dose: 2-4 units, titrate 1 unit/day to target CBG

(if on SUs, meglitinidies or TZDs, consider stopping or adjusting due to hypoglycemia risk)

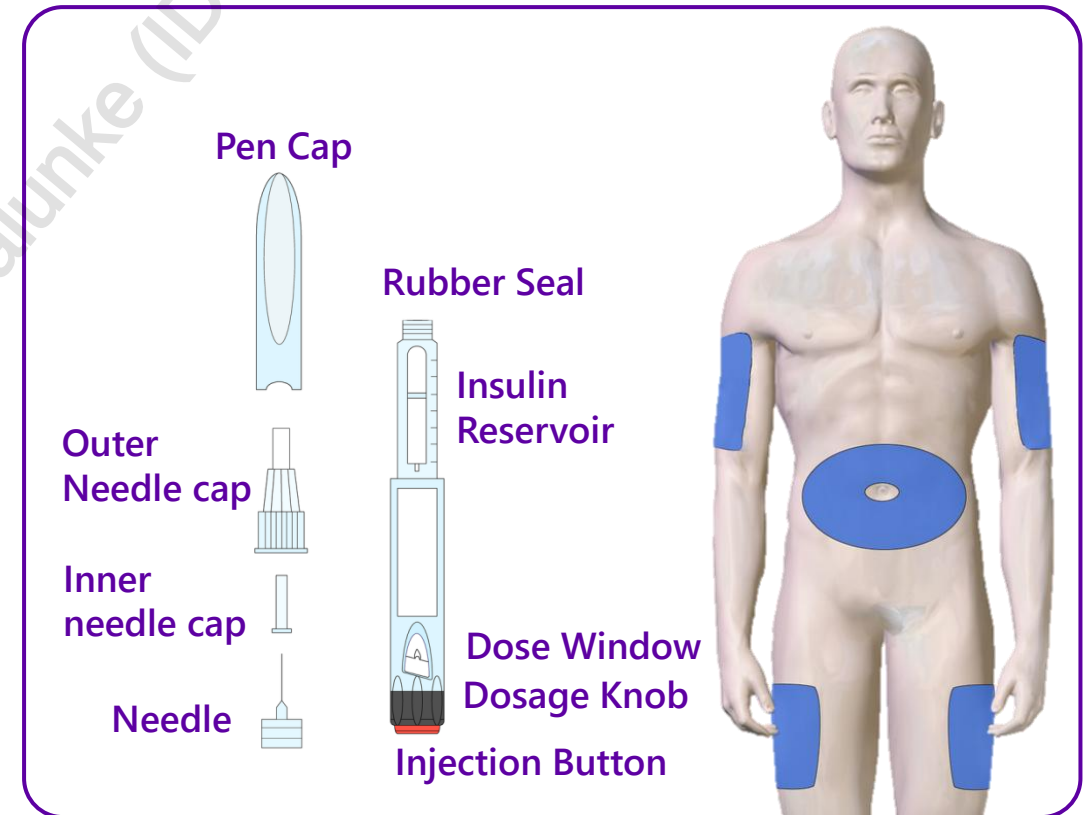
INSULIN ADMINISTRATION

PEN

1. Wash your hands
2. Check the expiration date of the pen
3. Remove pen cap and wipe tip of pen with alcohol swab
4. Put needle onto pen; remove plastic outer and inner needle cap
5. Perform an **air shot of at least 2 units** to clear bubbles
6. Dial dose to the number of units required
7. **Pinch a fatty area to inject***, insert needle at a 45-90 degree angle and push plunger
8. **Hold for about 10 seconds**, then withdraw
9. Replace outer needle cap and twist needle off pen
10. **Dispose needle into a sharps bin**

*Possible injection sites: **outer thighs, back of arms, abdomen (preferred), buttocks**

Ensure that **injection sites are rotated** to avoid "lumpy" skin



Store **unopened pens** in the refrigerator between **2-8°C**. Once **used**, store pens at **room temperature** for up to 28 days.

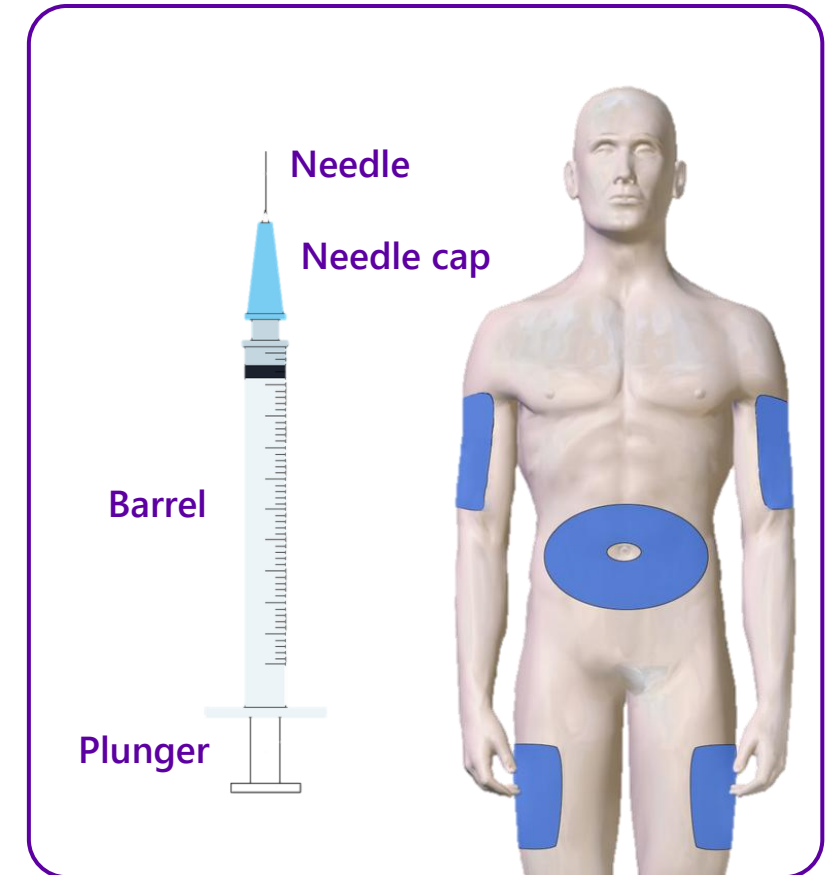
INSULIN ADMINISTRATION

SYRINGE

1. Wash your hands
2. Check the expiration date of the vial and remove lid
3. Wipe rubber top and injection site with alcohol swab
4. Pull needle cap off. Pull back plunger to number of units required
5. Push needle through the rubber top of the vial. Push down plunger to push air into vial
6. Pull the plunger back until you reach number of units required
7. Remove any air bubbles in syringe by tapping the barrel
8. Pinch a fatty area to inject*, insert needle at a 45-90 degree angle and push plunger
9. Remove needle from skin
10. Dispose needle into a sharps bin

*Possible injection sites: outer thighs, back of arms, abdomen (preferred), buttocks

Ensure that injection sites are rotated to avoid "lumpy" skin



SOMOGYI EFFECT AND DAWN PHENOMENON

Somogyi Effect

Nocturnal Hypoglycemia (3AM) → body reacts with **rebound hyperglycemia**

- Check BG around 3 am for multiple nights & again upon awaking
- Eating a snack at bedtime may help
- Decreasing insulin in the evening may help

Dawn Phenomenon

Early (3-8AM) release of hormones → ↑ **BG (no nocturnal hypoglycemia)**

- Check BG around 3 am for multiple nights & again upon awaking to **rule out Somogyi effect**
- Avoiding snacks at bedtime may help
- Eating dinner earlier or exercising after dinner may help
- Taking insulin closer to bedtime may help

DIABETIC KETOACIDOSIS

An acute and **fatal metabolic complication** of DM due to **severe insulin deficiency**

Characteristics

- Variable **hyperglycemia**
- Volume depletion, acidosis
- **↓ consciousness**
- Ketones in urine and/or blood
- **Depleted Na⁺, K⁺, Cl⁻, H₂O**
- **Most common in T1DM**
- May be seen in new pts or **pts nonadherent to insulin**
- Associated with **onset of severe illness**

Management

1. Hospitalization to ICU
2. Fluid resuscitation and K⁺ replacement
 - Check Na⁺ and K⁺ q2h
 - Guidelines advise giving K⁺ if levels are <5.5mmol/L (without delaying insulin, unless K⁺ < 3.3)
 - Insulin and dextrose infusion
 - IV insulin until BG is 12-14 mmol/L, check BG q1h
4. Bicarbonate
 - **Beneficial when pH < 7.0**; measure levels q2h
5. Treat underlying cause (e.g. infection or non-adherence)

VASCULAR PROTECTION

SCENARIO	In addition to nonpharmacological measures: exercise, healthy diet, and smoking cessation
Statin	• For documented macrovascular disease • Age ≥ 40 for T2DM • Age < 40: • DM for ≥ 15 years & age ≥ 30 • Microvascular complications • Therapy warranted based on Canadian Cardiovascular Society Lipid Guidelines
ACEi/ARB	• Diabetes and HTN with CVD or CKD, including microalbuminuria, or with CV risk factors $\rightarrow$ first line choices are ACEi or ARB • Diabetes and HTN without factors listed above $\rightarrow$ choices are ACEi, ARB, DHP-CCB, thiazide/thiazide-like diuretics
Aspirin[®] (low dose ASA 81-162 mg)	• Do not routinely use for primary prevention in patients with DM • Use for secondary prevention of CVD (i.e. history of MI/stroke)
GLP-1 Agonist or SGLT2 Inhibitor	• Reduce MACE: liraglutide, dulaglutide, sc semaglutide, empagliflozin, canagliflozin • Reduce HF hospitalization: dapagliflozin, empagliflozin • Protect kidney: canagliflozin, dapagliflozin, empagliflozin

DIABETIC NEUROPATHY

Nerve damage (peripheral/autonomic/motor) due to chronically uncontrolled hyperglycemia

Characteristics

- Hallmark symptoms:
 - Loss of sensations in toes and feet
 - Presence of sharp shooting pains, burning/tingling sensation, throbbing/numbness sensation
- Nerves of the legs/feet most affected (advanced → can progress to hands/arms)
- Can also affect...
 - Autonomic nerves: bowel/bladder problems, trouble swallowing, dizziness, excessive sweating, sexual dysfunction
 - Motor nerves: muscle weakness/wasting leading to claw toes, hammer toes, high-arched foot, or flat foot
- Increases risk for foot ulcers and amputation

Management

- **Prevent:** blood glucose control
- **Screen annually:** 10g monofilament test or vibration perception
- **Manage risk factors:** elevated blood glucose, elevated triglycerides, high BMI, smoking, and hypertension
- **Care for Feet:**
 - Examine feet daily (skin changes/structural abnormalities)
 - Engage in regular nail care (cut nails straight across)
 - Apply moisturizing lotion if feet are dry (not between the toes)
 - Wear properly fitted footwear
 - Avoid extreme cold/heat
- **Treat nerve pain:** antidepressants (e.g. amitriptyline, duloxetine, venlafaxine), anticonvulsants (e.g. pregabalin, gabapentin, valproate), nitrate spray, capsaicin cream, TENS, opioids* (*last line)

SICK DAY MANAGEMENT

S	Sugar	• Measure blood sugar more frequently (q3-4h and at bedtime)
I	Insulin	• Always take your basal insulin → noncompliance can lead to DKA • Use short acting insulin if blood sugar is elevated
C	Carbohydrates	• Maintain fluids (with minimal sugar), mineral and carbohydrate intake • Limit caffeine
K	Ketones	• Check urine ketones q3-4h and at bedtime



Medications to **hold**: **SADMANS**

Sulfonylureas, **A**CEi, **D**iuretics, **M**etformin, **A**RBs, **N**SAIDs, **S**GLT2 inhibitors

SICK DAY MANAGEMENT

Guide patients to contact their Health Care Team and/or go to the Emergency Department:

- If they cannot drink enough fluids
- If they does not know which medications to stop
- If they does not know how to adjust insulin
- If blood glucose stays higher than 14 mmol/L for 2 consecutive readings and does not respond to increased insulin
- If urine ketones are moderate to high
- If blood glucose stays lower than 4 mmol/L for 2 consecutive readings and does not respond to hypoglycemia treatment
- If they have any of the following that are not getting better:
 - Vomiting, diarrhea, stomach pain,
 - Frequent urination, extreme thirst,
 - Weakness,
 - Difficulty breathing or
 - Fever

SPECIAL POPULATIONS - PREGNANCY

Pre-pregnancy

- **Folic acid supplementation** (3 months prior to conception)
- **Eye exam** (pregnancy can accelerate retinopathy due to poor glycemic control)
- Switch to an **insulin regimen** (target **HbA1c: $\leq 7\%$ or $\leq 6.5\%$ if safely achievable**)
- **Screen for CVD and CKD**
- Discontinue any teratogenic medications (i.e. ACEi, ARB, statin)

Pregnancy

- **Rapid acting insulin** or **regular insulin**
 - **Lispro** and **aspart** have **decreased risk of hypoglycemia** compared to regular insulin
 - Patients with pre-existing diabetes: NPH insulin, detemir, or glargine can be used
- If insulin is contraindicated, **metformin or glyburide can be used (off-label indication)**
- If patients already have DM, continue insulin, metformin, and/or glyburide if already taking
- HbA1c: $\leq 6.5\%$ or $\leq 6.1\%$ (if safely achievable)
- FPG: < 5.3 mmol/L (non-pregnant: 4-7 mmol/L)
- 1-hour PPG: < 7.8 mmol/L
- 2-hour PPG: < 6.7 mmol/L (non-pregnant: 5-10 mmol/L)

BREASTFEEDING

- **Insulin is secreted in the breastmilk** but poses little risk to the infant (degraded in the GI tract)
- Glyburide and metformin have been used but **no long terms studies** have been done
- **Breastfeeding increases risk of hypoglycemia** due to increased caloric requirements
- Frequent blood glucose monitoring is crucial
- Gestational diabetes
 - In many cases, hyperglycemia will completely resolve
 - All antihyperglycemics will need to be discontinued
 - Should have a 75 g OGTT between 6 weeks and 6 months following delivery to detect persistent hyperglycemia

SELF MONITORING OF BLOOD GLUCOSE

Capillary blood glucose testing (most common)

- Continuous glucose monitoring (can be part of an insulin pump)
- Flash glucose monitoring (e.g. FreeStyle Libre®, Dexcom®, Medtronic Guardian® with Medtronic pump)

Patients on insulin

- **Basal + bolus regimen - minimum of 3 times per day** and include pre and post prandial measurements
- **Basal insulin + oral antihyperglycemic agents – at least once a day** at variable times

Patients NOT on insulin

- Individualize based on type of anti-hyperglycemic agents, level of control and risk of hyperglycemia. For example:
 - If A1C is at target or receiving anti-hyperglycemic agents not associated with hypoglycemia → daily CBG testing isn't required or recommended, unless pt is unwell/risk of ↑BG
 - If A1C is not at target → initiate structured CBG testing (including a 7-point profile)

GDM/T2DM during pregnancy

- If on no antihyperglycemics → test CBG 4x on alternate days
- If receiving insulin therapy → test CBG 4x/day (fasting & postprandial)

VACCINATION GUIDELINES FOR ADULTS LIVING WITH DIABETES

Recommendation level	Ongoing immunizations	One-time vaccinations series
Strong	1. COVID-19: at least 3 months from last dose 2. Flu Shot: annually 3. Tetanus, diphtheria, pertussis: booster every 10 years	1. Regardless of pneumococcal 13 or 23 status, administer one dose of pneumococcal 20 or 21 in patients ≥ 65 years old 2. Administer one dose of recombinant zoster vaccine in patients ≥ 50 years old. If the patient received a live zoster vaccine or had an episode of herpes zoster, wait at least 1 year 3. Hepatitis B if not vaccinated during childhood or if there is a change in medical condition* 4. RSV: in patients ≥ 75 years old

*e.g. patient presents with CKD with or without dialysis. Most patients should be immunized with hepatitis A and B

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
SMBG	FBG: 4-7 mmol/L PPG: 5-10 mmol/L	Ongoing
HbA1c	≤ 7%*	At 3 months, then ongoing
BP	<130/80 mmHg	3-6 months, then ongoing
LDL	<2 OR ↓ by 50%	Ongoing
Urine Albumin to Creatinine ratio (ACR)	<2 mg/mmol	Ongoing

*Lower or higher A1C target may be appropriate for some individuals.

MONITORING PARAMETERS

Safety Endpoints		
Medication AEs	Prevent or minimize	Ongoing
Screening for complications	Cardiac	BP measurements at all visits, lipid profile at time of diagnosis and then yearly if normal
	Foot	Monofilament/vibration yearly or more if abnormal
	Kidney	eGFR and ACR yearly, or more if abnormal
	Retinopathy	annually (Type 1), every 1-2 years (Type 2)

TAKEAWAY

- First-line treatment Type 1 diabetes: insulin (basal/bolus)
- First-line treatment Type 2 diabetes: metformin (for ALL patients)
 - Add-on agent preference depends on patient-specific factors
 - Metabolic decompensation (marked hyperglycemia, ketosis, etc.) → insulin
 - High CV risk → GLP1-RA or SGLT2i
 - HF → SGLT2i
 - CKD → SGLT2i, may consider GLP1-RA
 - None of the above → consider individual patient factors and goals (weight loss, cost, etc.)
- Screen and manage microvascular and macrovascular complications
- Sick day management: hold SADMANS
- Targets:
 - FPG (mmol/L): 4.0-7.0 (to achieve A1C less $\leq 7\%$)
 - 2 hr-post-prandial (mmol/L): 5.0-10.0 (if A1C is at target level) or 5.0-8.0 if A1C not at target
 - HbA1c: $\leq 7\%$ (or $\leq 8.5\%^*$) or $\leq 6.5\%$ in T2DM with low hypoglycemia risk to reduce CKD & retinopathy risk

CASE 1

SM is a 45-year-old female who was diagnosed with Type 2 diabetes mellitus by her family physician. Her doctor phones you to discuss her care. SM lives independently and does not have any functional limitations. She denies any symptoms of hyperglycemia. Her past medical history is significant for osteoarthritis, asthma, hypertension and a prior stroke for which she did not suffer any deficits. She takes Advair® (fluticasone propionate and salmeterol) 250/50 mcg MDI – 1 puff PO BID, ASA 81 mg PO daily, Tylenol® (acetaminophen) PRN and ramipril 10 mg PO daily. She is not allergic to anything and quit smoking 10 years ago. Her A1C is 8.9%, eGFR is 60 mL/min, BP 142/84 mmHg, HR 84 BPM.

1. What is an appropriate A1C target for SM?

- a) 6 %
- b) 7 %
- c) 8 %
- d) 8.5 %



CASE 1

SM is a 45-year-old female who was diagnosed with Type 2 diabetes mellitus by her family physician. Her doctor phones you to discuss her care. SM lives independently and does not have any functional limitations. She denies any symptoms of hyperglycemia. Her past medical history is significant for osteoarthritis, Chronic Obstructive Pulmonary Disease (COPD), hypertension and a prior stroke for which she did not suffer any deficits. She takes Advair® (fluticasone propionate and salmeterol) 250/50 mcg MDI – 1 puff PO BID, ASA 81 mg PO daily, Tylenol® (acetaminophen) PRN and ramipril 10 mg PO daily. She is not allergic to anything and quit smoking 10 years ago. Her A1C is 8.9%, eGFR is 60 mL/min, BP 142/84 mmHg, HR 84 BPM.

2. What therapy should be initiated at this point?

- a) Empagliflozin and liraglutide
- b) Metformin and sitagliptin
- c) Insulin
- d) Metformin and empagliflozin



CASE 2

MD is a 65-year-old female who has had diabetes for 6 years. Recently, she has been feeling lightheaded and unwell in the mid-afternoon, right after she goes to the gym. Her past medical history is significant for hypertension, Type 2 diabetes and dyslipidemia. She takes ramipril 10mg PO daily, hydrochlorothiazide 25mg PO daily, metformin 1000mg BID and gliclazide MR 120mg PO daily. She is not allergic to anything. Her A1C is 6.7%, eGFR is 70 mL/min, LDL is 1.98 mmol/L, BP 128/75 mmHg, HR 84 BPM.

1. What would you recommend for MD?
 - a) Tell her to change her exercise time
 - b) Ask her to check her blood pressure after she goes to the gym
 - c) Ask her to check her blood sugar after she goes to the gym
 - d) Ask her to check her blood sugar before she goes to the gym



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MD is a 65-year-old female who has had diabetes for 6 years. Recently, she has been feeling lightheaded and unwell in the mid-afternoon, right after she goes to the gym. Her past medical history is significant for hypertension, Type 2 diabetes and dyslipidemia. She takes ramipril 10mg PO daily, hydrochlorothiazide 25mg PO daily, metformin 1000mg BID and gliclazide MR 120mg PO daily. She is not allergic to anything. Her A1C is 6.7%, eGFR is 70 mL/min, LDL is 1.98 mmol/L, BP 128/75 mmHg, HR 84 BPM.

2. She notices that her blood sugar is 3.8 when she feels unwell, what would you suggest to her family MD?
- a) Reduce metformin to 500mg BID
 - b) Reduce gliclazide MR to 60mg daily
 - c) Do not change medications, ask her to eat meals before going to the gym
 - d) All the above are appropriate options



CASE 2

MD is a 65-year-old female who has had diabetes for 6 years. Recently, she has been feeling lightheaded and unwell in the mid-afternoon, right after she goes to the gym. Her past medical history is significant for hypertension, Type 2 diabetes and dyslipidemia. She takes ramipril 10mg PO daily, hydrochlorothiazide 25mg PO daily, metformin 1000mg BID and gliclazide MR 120mg PO daily. She is not allergic to anything. Her A1C is 6.7%, eGFR is 70 mL/min, LDL is 1.98 mmol/L, BP 128/75 mmHg, HR 84 BPM.

3. Six months later, MD reports weight gain of 4 kg despite the gym and inquires if her diabetes medications can be optimized further in any way?
- a) Stop metformin and start subcutaneous dulaglutide
 - b) Stop gliclazide and start subcutaneous dulaglutide
 - c) Stop gliclazide and start sitagliptin
 - d) MD's medications are already optimized



CASE 3

JP is a 56-year-old male who works as a marketing executive in the city. His past medical history is significant for Type 2 diabetes, GERD and hypertension. He takes ramipril 2.5mg PO daily, rosuvastatin 20mg PO daily, insulin lispro 6 units at breakfast, 8 units at lunch and 13 units at dinner and insulin glargine 2 units at bedtime. His fasting blood glucose target is 4 to 7mmol/L. His food intake and activity levels are consistent and he is not allergic to anything. His current HbA1c is 7.3%.

Before Breakfast	Before Lunch	Before Dinner	Bedtime
8.1	5.6	6.0	10.1
5.7	7.2	8.0	9.2
4.9	6.3	6.7	10.5

1. What dose adjustment would you suggest based on his readings above?
 - a) Decrease breakfast rapid-acting insulin by 1-2 units
 - b) Increase lunch rapid-acting insulin by 1-2 units
 - c) Increase dinner rapid-acting insulin by 1-2 units
 - d) Increase bedtime long-acting insulin by 1-2 units



CASE 3

JP is a 56-year-old male who works as a marketing executive in the city. His past medical history is significant for Type 2 diabetes, GERD and hypertension. He takes ramipril 2.5mg PO daily, rosuvastatin 20mg PO daily, insulin lispro 6 units at breakfast, 8 units at lunch and 13 units at dinner and insulin glargine 2 units at bedtime. His fasting blood glucose target is 4 to 7mmol/L. His food intake and activity levels are consistent and he is not allergic to anything. His current HbA1c is 7.3%.

Before Breakfast	Before Lunch	Before Dinner	Bedtime
8.1	5.6	6.0	10.1
5.7	7.2	8.0	9.2
4.9	6.3	6.7	10.5

2. JP plans to go on vacation and will have inconsistent mealtimes. He wonders how to manage missed meals.
- a) No change to regimen
 - b) Skip bedtime long-acting insulin dose
 - c) Decrease missed bolus insulin dose(s) by 50%
 - d) Skip missed bolus insulin dose(s)



BONUS PRACTICAL CASE EXAMPLE

JT is a 14-year-old female who has just been diagnosed with type 1 diabetes. She states she has been going to the washroom more than usual, has a thirst that cannot be quenched with liquids, and her breath smells like nail polish remover. Her A1C upon today's visit was 9.3% and her fasting BG was 8.2 mmol/L. She has a remarkable medical history otherwise. Her MD would like to start her on 10 units of insulin aspart prior to each meal and 30 units of glargine insulin each night before bed. Insulin aspart is available in a concentration of 100 U/mL. Each insulin aspart pen has a volume of 3 mL and each box of insulin aspart contains 5 pens. How many pens does JT need for a 28-day supply of her bolus insulin?

- a) 4
- b) 2
- c) 1
- d) 3



BONUS PRACTICAL CASE EXAMPLE

TD is a 68-year-old with type 2 diabetes, hypertension, CKD and dyslipidemia. He is currently taking ramipril 20mg PO OD, atorvastatin 80mg PO OD and insulin NPH 15 units BID in the morning and at bedtime. You receive a call from his physician asking for advice about switching the patient to a long-acting insulin as they would no longer like to take multiple daily injections. Calculate the new dose of insulin the patient should receive if they are being switched to once-daily dosing of insulin detemir.

- a) 15 units once daily
- b) 20 units once daily
- c) 24 units once daily
- d) 30 units once daily



FREQUENTLY ASKED QUESTIONS

1. How many times do you retreat for hypoglycemia?

- Until the blood glucose is > 4 mmol/L.

2. Metformin is not to be used in End Stage Renal Disease (ESRD). But according to the algorithm it would be metformin + 2nd agent. Do we remove the metformin and choose one of the agents safe in ESRD?

- Yes, if the patient is in ESRD, metformin should not be started. Monotherapy is unlikely to reduce A1C by 1.5% or more, thus starting a second agent would have added benefit. Note: The diabetes guidelines do not have a concrete answer regarding this

3. When can metformin be used in chronic kidney disease?

- eGFR $\geq$ 45 mL/min: No dosage adjustment necessary
- eGFR 30 mL/min to 44 mL/min: Initial therapy with 500 mg once daily with the evening meal titrated to 500 mg twice daily, if tolerated and w/ close monitoring of kidney function
- eGFR 44 mL/min to 15 mL/min: Maximum daily dose is 500 mg, if tolerated and with close monitoring of kidney function
- eGFR < 15 mL/min Use is contraindicated

4. When are mixed insulins recommended/best used?

- To provide simple insulin regimens for patients e.g. 2 injections/day versus 4. You may see it more commonly in older populations.

5. Is insulin always the first option for treatment of type 2 diabetes mellitus with symptomatic hyperglycemia?

- Yes, for adults with type 2 diabetes mellitus and metabolic decompensation (e.g. marked/symptomatic hyperglycemia, ketosis or unintentional weight loss), insulin should be used (recall: metabolic decompensation may include dehydration, diabetic ketoacidosis, hyperglycemic state)

FREQUENTLY ASKED QUESTIONS

8. Insulin: Is the priming of the pen/air shot required only when a new pen is started or every time before an injection?

- Prime before each use

9. Insulin: Novolin NPH BID, now also receiving Humalog TID before meals. How often should the patient test their blood glucose?

- For any individual taking insulin more than once daily, test min 3x/day and include both pre/post prandial measurements.

10. Are ramipril, perindopril, telmisartan the only ACEi/ARB to be used in diabetes mellitus?

- You want to use ACEi/ARBs at doses that have proven vascular protection e.g. telmisartan 80 mg. The ones mentioned are listed in the guidelines. If there are other agents with proven vascular protection those may also be considered.

11. Site of insulin injection for women in pregnancy?

- Outer thighs, backs of the upper arms

12. What are the reasons insulin is contraindicated in pregnancy?

- Insulin is not contraindicated in pregnancy, it is safe and does not cross the placenta

13. In patients with diabetic ketoacidosis, if the potassium level is less than 3.3, do we give insulin?

- No, correct the potassium first then administer insulin

14. What is the dose of insulin given to patients with diabetic ketoacidosis?

- IV insulin at 0.1 units/kg/hour, infusion ongoing until ketosis is resolved. Start IV dextrose to avoid hypoglycemia.

15. What is dysglycemia? Is it the same as diabetes? What causes it?

- Dysglycemia refers to abnormal serum glucose levels. It can be caused by medications, pancreatic diseases, autoimmune conditions, Cushing's syndrome, severe infections, stress, and hormone-secreting tumours. Dysglycemia is not the same as diabetes. Dysglycemia may or may not be caused by diabetes. Examples of medications that can induce dysglycemia include beta-blockers (e.g. atenolol), corticosteroids (e.g. prednisone), immunosuppressants (e.g. tacrolimus), isoniazid, niacin, protease inhibitors (e.g. ritonavir), thiazide and loop diuretics (e.g. chlorthalidone, furosemide), second-generation antipsychotics (e.g. olanzapine, quetiapine, clozapine), pentamidine, and pasireotide.

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CHANGE LOG

May 2025

- Slide 3: Removed UTI from legend
- Slide 19: Updated eGFR C/I recommendation for metformin
- Slide 24: Removed UTI risk with SGLT2i and added 'genital mycotic infection'
- Slide 24: Updated renal info for SGLT2i
- Slide 37: Updated insulin starting dose recommendation
- Slide 61: Updated metformin FAQs

July 2025

- Formatting and grammar changes throughout
- Slide 9: Emphasized what gestational diabetes is
- Slide 30: Added regular insulin is available IV and used in DKA/HHS
- Slide 35: Added T1DM many patients use insulin pumps
- Slide 46: Added to continue certain meds if pre-existing DM and becomes pregnant
- Slide 48: Added Medtronic Guardian to CGMs

October 2025

- Slide 19: Added note on metformin use in renal impairment
- Slide 24: Adjusted renal dose reduction cutoffs for SGLT2 inhibitors as per Diabetes Canada
- Slide 49: Added vaccination recommendation for patients living with Diabetes
- Slide 64: Updated references

November/December 2025

- Slide 15: Added that any amount of alcohol consumption is harmful
- Slide 19: Changed crcl in footnote to eGFR to match guidelines
- Slide 24: Updated to say that empagliflozin, canagliflozin and dapagliflozin slow *nephropathy* progression

February 2026

- Slide 2: Updated NAPRA competencies
- Minor phrasing changes throughout
- Slide 37: Moved to insulin section to introduce basal/bolus concepts prior to talking about their application
- Slide 41: Updated DKA management
- Slide 48: Elaborated on 'Patients NOT on insulin' section
- Slide 50: Updated timeframe of HbA1C